ADITYA SAXENA

◇ aditya.saxena2511@gmail.com ◇ +91-9711366368 ◇ GitHub ◇ LinkedIn

Education

Indian Institute of Technology Delhi **Delhi, India**
BTech in Engineering & Computational Mechanics, Minor in Computer Science *May 2026*
GPA: 8.85/10.0 (Dept Rank 5)
Awards: Summer Undergraduate Research Award (2024), Semester Merit Award (Sem-II 23-24)
Selected Coursework: NLP, Advanced Reinforcement Learning, Probabilistic ML, Database Management Systems, Information retrieval and Web Search, ML, Digital Image Processing, Analysis and Design of Algorithms, Discrete Mathematics, Probability and Stochastic Processes, Linear Algebra & Differential Equations

Technical Skills

- **Programming Languages** - Python, C++, Javascript
- **ML/DL Libraries:** PyTorch, OpenCV, NumPy, Pandas, Matplotlib
- **Frameworks/Tools:** Huggingface, Gymnasium, MongoDB, Express.js, React.js, Node.js, Git

Professional Experience

IBM Research Labs — (AI Research Intern) **May 2025 – July 2025**

- Implemented a state-driven multi-agent LLM workflow using Finite State Machines (FSMs) for robust orchestration of Earth Observation tasks, improving execution reliability over single-agent baselines.
- Designed a modular LoRA-based SFT pipeline for the orchestrator and specialist agents to learn state transitions and structured tool invocation, improving correctness rate and success rate by 12-15%.
- **Best Code Repository** Runners Up award during internship at IBM Research Labs.

Projects

Multilingual CoT Distillation **April 2026**

- Distilled CoT reasoning from Qwen2.5-7B into Qwen2.5-1.5B and Llama-3.2-1B students across 5 languages.
- Countered Indic token fertility with English-pivot self-consistency (k=4) + target-language rendering; raised usable Tamil/Kannada trace yield by 20% via STaR-style gold-conditioned rationalization.
- Eliminated an 9× gradient skew towards Indic scripts with per-example length normalization and added script-adaptation pretraining to reduce byte-fallback corruption.

Cross-Lingual Relation Extraction with Decoder-Only LMs **March 2026**

- Built a LoRA-tuned Qwen2.5-1.5B relation classifier for 5 languages by pooling query-conditioned final-token and entity-marker states. Used a staged curriculum of language-adaptive pretraining on unlabelled Indic corpora followed by task tuning on upsampled Indic data.
- Reframed generative extraction as label-ID prediction under greedy decoding to eliminate hallucinated labels, trained with interleaved LM and task loss to gain coverage of all four Indian languages.
- Designed a training-free in-context learning pipeline on Llama-3.1-8B using a two-block prompt with a byte-identical ontology prefix for vLLM prefix caching.

Training & Sampling from DDPM **March 2026**

- Implemented DDPM and DDIM from scratch in PyTorch, incorporating sinusoidal time embeddings, residual denoising network. Conducted quantitative evaluation across variance schedules using mode coverage, Mahalanobis-distance analysis, and deterministic DDIM sampling with varying inference steps.

Decoder-only Transformer **Dec 2025**

- Implemented a **decoder-only Transformer** from scratch in PyTorch, including **multi-head causal self-attention, sinusoidal positional encodings, residual connections**, and **LayerNorm.**
- Trained on TinyStories Dataset using FastTest embeddings, **mixed-precision (AMP), gradient accumulation**, **cosine LR schedule** and **perplexity-based evaluation.**
- Built an **efficient autoregressive generation** pipeline with **KV-cache, temperature/top-k sampling**, and **beam search**, enabling fast text decoding and coherent story generation.